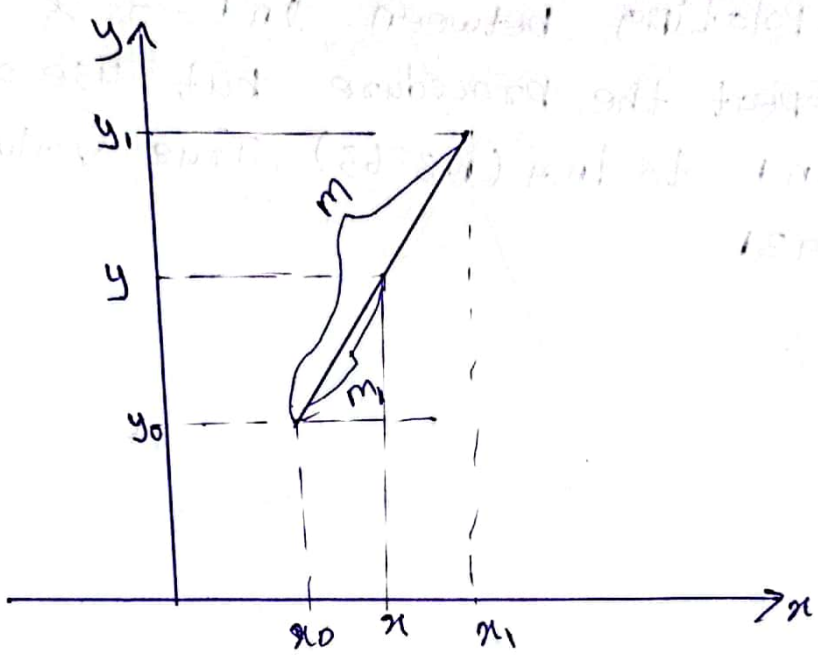
Interpolation

Linear Interpolation :



m and m_1 equal

So we can write:-

$$m_1 = m$$

$$\frac{y - y_0}{x - x_0} = \frac{y_1 - y_0}{x_1 - x_0}$$

$$y - y_0 = \frac{(y_1 - y_0)}{(x_1 - x_0)} (x - x_0)$$

$$y = y_0 + \frac{(y - y_0)}{(x_1 - x_0)} (x - x_0).$$

Ex :-

Given

$(x_0, y_0) = (1, 3)$

$(x_1, y_1) = (4, 15)$

Find y when $x = 2$

$$y = 3 + \frac{(x-1)(15-3)}{4-1}$$

$$= 3 + \frac{1 \cdot 12}{3}$$

$$= 3 + 4$$

$$= 7$$

So, the interpolated value at $x = 2$ is 7

Lagrange Interpolation

$$y = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} y_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} y_2$$

given

$x = (1, 3, 2)$

$y = (6, 4, 5)$

$x = 0$

3 data points

so this is second order

lagrange Interplation

$$y = \frac{(1-3)(1-2)}{(1-3)(1-2)} \times 6 + 0 + 0$$

$$= \frac{-2x-1}{-2x-1} \times 6$$

$$= 6.$$

Newton's Interpolating Polynomials

$$f_n(x) = b_0 + b_1(x-x_0) + b_2(x-x_0)(x_0-x_1) + \cdots b_n(x-x_0)\cdot(x-(x_{n-1}))$$

$b_0 = F(x_0)$

$b_1 = F(x_0, x_1)$

$\vdots$

$b_n = F(x_0, x_1 \cdots x_n)$

we can write

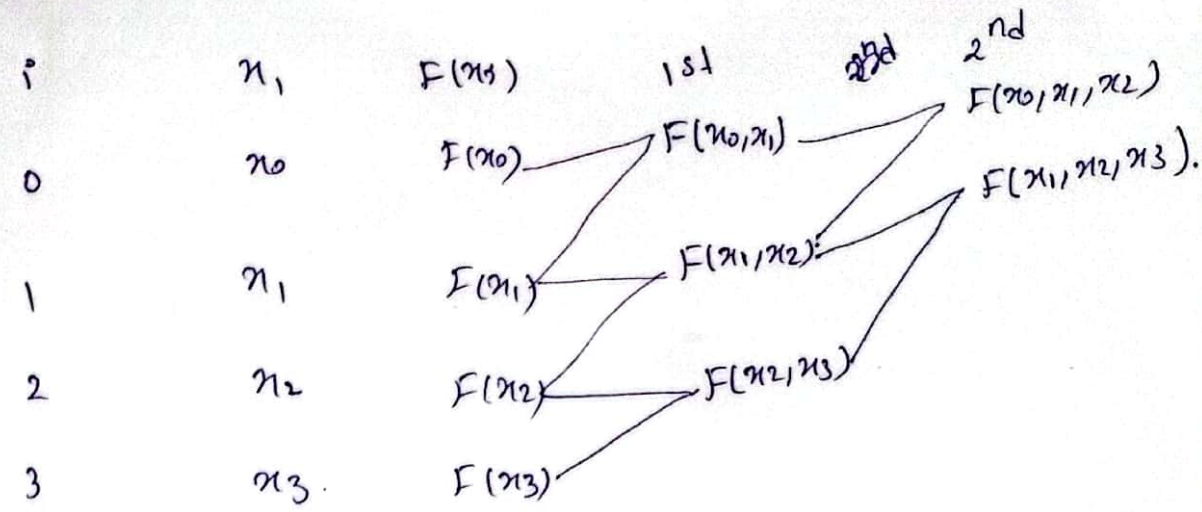
$$F(x_0, x_1) = \frac{F(x_0) - F(x_1)}{x_0 - x_1}$$

$$b_2 = F[x_0, x_1, x_2)$$

$$= \frac{F(x_0, x_1) - F(x_1, x_2)}{x_0 - x_2}.$$

similarly, we can write n^{th} Finite divided differece is

$$F(x_0, x_1, x_2 \cdots x_n) = \frac{F(x_0 \cdots x_{n-1}) - F(x_1 \cdots x_n)}{x_0 - x_n}$$



This is divided difference table

Ex:- Given Data points

(1, 2) · (2, 3) · (4, 10) $x = 3$

d·Fdd table

x	$F(x)$	1st	2nd
1	2		
2	3	1	
4	10	3.5	0.833

$$F_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

$$= 2 + 1(3 - 1) + 0.833(3 - 1)(3 - 2)$$

$$= 2 + 2 + 0.833 \times 2$$

$$= 2(1 + 1 + 0.833) \quad = 2.833 \times 2$$

$$= 5.666$$